

Summary of work on the Magnus expansion

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We want to write a summary on the advancements we made on the integration with Magnus expansions.

In this document, we will start presenting the advantages of the numerical integration using the Magnus expansion for the orientation of the rod cross sections. We will then describe how we can extend the numerical integration in the case of linear, non-homogenous ODEs.

1 Consideration on performances

When integrating the ODEs for the IDM, but for the TIDM as well, we perform an in-cascade integration. Before beginning the integration, it is practical to evaluate the strains at every point used in the integration, and store the results in some vector.

With this practice, we compute all the strain only once and then they are available for all the integrators : Quaternion (or rotation matrix), position, angular velocities, ecc.

1.1 Reconstruction of the field of strains

We begin this section with some consideration on our task: compute the field of strain at the quadrature points place in between the Chebyshev points. For this aim, we recall the equations describing the field of strain of the rod.

$$\begin{aligned}\xi_{(X)} &= B\Phi_{(X)}q_{(t)} + \bar{B}\xi_c \\ \dot{\xi}_{(X)} &= B\Phi_{(X)}\dot{q}_{(t)} \\ \ddot{\xi}_{(X)} &= B\Phi_{(X)}\ddot{q}_{(t)}\end{aligned}\tag{1}$$

In this context, we first observe that, knowing the Chebyshev point we can define the quadrature points in the rod domain. This means that the map $B\Phi_{(X)}$ can be precomputed at all the points in initialisation and the stored into an appropriate vector. The algorithm 1 resumes the procedure, using a six order quadrature with Gauss-Legendre points as in [1].

```
 $B\Phi\_stack = []$   
 $h\_stack = []$ 
```

```
Compute the  $N_c$  Chebyshev points  
 $Chebyshev\_points \leftarrow computeChebyshevPoints(N_c)$ 
```

```
for  $i \in [0, N_c - 1)$  do
```

```
     $begin = Chebyshev\_points[(N_c - 1) - i]$   
     $end = Chebyshev\_points[(N_c - 1) - (i + 1)]$ 
```

```
     $h = end - begin$   
     $step = \sqrt{15}/10$ 
```

```
     $h\_stack[i] = h$ 
```

```
     $x_1 = begin + (0.5 - step) * h$   
     $x_2 = begin + 0.5 * h$   
     $x_3 = begin + (0.5 + step) * h$ 
```

```
     $B\Phi_{(x_1)} = B\Phi_{(x_1)}$   
     $B\Phi_{(x_2)} = B\Phi_{(x_2)}$   
     $B\Phi_{(x_3)} = B\Phi_{(x_3)}$   
     $B\Phi\_stack[i] = \{B\Phi_{(x_1)} \ B\Phi_{(x_2)} \ B\Phi_{(x_3)}\}$ 
```

```
end
```

Algorithm 1: The algorithm for computing the strains at every quadrature point

The $B\Phi_stack$ vector will then be used, from now on in the integration, to reconstruct the field of strain along the rod. We can now compute ξ , $\dot{\xi}$, $\ddot{\xi}$ at every quadrature point. Algorithm 2 shows the procedure.

The cost of algorithm 2 is shown in table 1, where we compare the time for computing all the strain variables ξ , $\dot{\xi}$, $\ddot{\xi}$ at the chebyshev points and at

```

 $\xi\_stack = []$ 
 $\dot{\xi}\_stack = []$ 
 $\ddot{\xi}\_stack = []$ 

for  $i \in [0, N_c - 1)$  do
     $\xi_1 = B\Phi\_stack[i]_{(x_1)}q + \bar{B}\xi_c$ 
     $\xi_2 = B\Phi\_stack[i]_{(x_2)}q + \bar{B}\xi_c$ 
     $\xi_3 = B\Phi\_stack[i]_{(x_3)}q + \bar{B}\xi_c$ 
     $\xi\_stack[i] = \{\xi_1 \ \xi_2 \ \xi_3\}$ 

     $\dot{\xi}_1 = B\Phi\_stack[i]_{(x_1)}\dot{q}$ 
     $\dot{\xi}_2 = B\Phi\_stack[i]_{(x_2)}\dot{q}$ 
     $\dot{\xi}_3 = B\Phi\_stack[i]_{(x_3)}\dot{q}$ 
     $\dot{\xi}\_stack[i] = \{\dot{\xi}_1 \ \dot{\xi}_2 \ \dot{\xi}_3\}$ 

     $\ddot{\xi}_1 = B\Phi\_stack[i]_{(x_1)}\ddot{q}$ 
     $\ddot{\xi}_2 = B\Phi\_stack[i]_{(x_2)}\ddot{q}$ 
     $\ddot{\xi}_3 = B\Phi\_stack[i]_{(x_3)}\ddot{q}$ 
     $\ddot{\xi}\_stack[i] = \{\ddot{\xi}_1 \ \ddot{\xi}_2 \ \ddot{\xi}_3\}$ 
end

```

Algorithm 2: The algorithm for computing the strains at every quadrature point

all the quadrature points.

From table 1, we see that reconstruct the strains, and their derivatives, at every quadrature point is more expensive than computing them at the Chebyshev points only. This drawback was expected as we need to perform 3 times more computations per Chebyshev point (using a quadrature of order 6). However, this part can be parallelised easily : we can create a thread pool and give at each thread some quadrature points where to compute the strains. This is possible as the computation of ξ , $\dot{\xi}$, $\ddot{\xi}$ depends only on X .

1.2 Integration with the Magnus expansions

Once the field of strains is reconstructed we can integrate using the Magnus expansion. We will show that, even in this part, we can take advantages of

$\xi, \dot{\xi}, \ddot{\xi}$	Time		
	$N_c = 11$	$N_c = 21$	$N_c = 31$
Chebyshev [μs]	0.857	1.69	2.68
Gauss-Legendre [μs]	3.61	7.22	10.6
Ratio	4.21	2.13	4.27

Table 1: Computational time need to compute ξ and derivatives on the Chebyshev and Gauss-Legendre points.

the shape of our problem. In the following, we consider the integration of the orientation of the rod cross section.

Once we compute all the strain field at all the quadrature points using the procedure explained in algorithm 2, we only need to compute $\Omega^{[6]}$ between every Chebyshev points. The algorithm 3 shows the integration using the Magnus expansion for the rotation matrix

The table 2 shows the differences, in computational time, of this optimised method with respect to *ode45* and the spectral integration.

From the table 2 we can make the following considerations

- The integration using *ode45* takes the same amount of time no matter of the number of points as it not in the Chebyshev grid. It seems that this is a valid option as its computational time remains stable. However, when we extend it to the full integration of variables in the IDM, its benefetis vanish.
- The spectral numerical integration suffers from the fact the number of Chebyshev points determines the dimension of the matrices. Having to perform a matrix inversion, the computational time grows exponentially with the number of points. The computational complexity is then $\mathcal{O}(N_c^3)$.
- The integration with Magnus expansion is incredibly efficient as it is composed of a series of multiplication of matrices of small dimension. Moreover, the number of points influences the total number of multiplications. For this reason the computational time grows linearly with the number of points. The computational complexity is then $\mathcal{O}(N_c)$.

It is then clear that the integration using the Magnus expansion is a valuable solution to speedup the IDM computations.

Moreover, if we consider the algorithm 3, we observe that the computation of $e^{\Omega^{[6]}}$ are completely indepent so we could as well parallelise the loop with a thread pool in oder to further reduce the computational time required to perform the integration.

$$stack_e^{\Omega^{[6]}} = []$$

```

for  $i \in [0, N_c - 1)$  do
     $[K_1 \ K_2 \ K_3] \leftarrow \text{Get curvature at quadrature point}$ 

     $A_1 = \hat{K}_1$ 
     $A_2 = \hat{K}_2$ 
     $A_3 = \hat{K}_3$ 

     $\alpha_1 = h\_stack[i] * A_2$ 
     $\alpha_2 = \frac{\sqrt{15}}{3}(A_3 - A_1)h\_stack[i]$ 
     $\alpha_3 = \frac{10}{3}(A_3 - 2A_2 + A_1)h\_stack[i]$ 

     $C_1 = [\alpha_1, \alpha_2]$ 
     $C_1 = -\frac{1}{60} [\alpha_1, 2\alpha_3 + C_1]$ 

     $\Omega^{[6]} = \alpha_1 + \frac{1}{12}\alpha_3 + \frac{1}{240} [-20\alpha_1 - \alpha_3 + C_1, \alpha_2 + C_2]$ 

    Stack the exponential into a vector
     $stack\_e^{\Omega^{[6]}}[i] = e^{\Omega^{[6]}}$ 
end

```

$$R_stack = []$$

$$R_stack[0] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

```

for  $i \in [0, N_c - 1)$  do
     $R\_stack[i + 1] = R\_stack[i] * stack\_e^{\Omega^{[6]}}[i]$ 
end

```

Algorithm 3: The algorithm for the Magnus integrator

Integration mode	Time		
	$N_c = 11$	$N_c = 21$	$N_c = 31$
<i>ode45</i> [μs]	148	141	139
Spectral [μs]	21.2	144	449
Magnus [μs]	1.65	3.42	5.01

Table 2: Computational time for reconstructing the rod orientation using *ode45*, spectral integration and the Magnus expansions.

2 Extension to non-homogenous ODE

We then discovered that the Magnus integrator can be extended to the case of an non-homogeneous ODE of type

$$y' = A_{(X)}y + b_{(X)} \quad (2)$$

Following the procedure explained in [2], it suffices to perform a change of variable to have

$$\begin{aligned} z' &= H_{(X)}z, \quad z = \begin{bmatrix} y \\ 1 \end{bmatrix} \\ \begin{bmatrix} y \\ 1 \end{bmatrix}' &= \begin{bmatrix} A_{(X)} & b_{(X)} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y \\ 1 \end{bmatrix} \end{aligned} \quad (3)$$

This allows us to implement the same formulation as before. It is worth noting that errors may occur when $\|A\| \ll \|b\|$. The occurrence of this situation should be tested; but a formulation exists to handle this case.

References

- [1] S. Blanes, F. Casas, J.-A. Oteo, and J. Ros, “The magnus expansion and some of its applications,” *Physics reports*, vol. 470, no. 5-6, pp. 151–238, 2009.
- [2] S. Blanes and E. Ponsoda, “Time-averaging and exponential integrators for non-homogeneous linear ivps and bvps,” *Applied Numerical Mathematics*, vol. 62, no. 8, pp. 875–894, 2012.